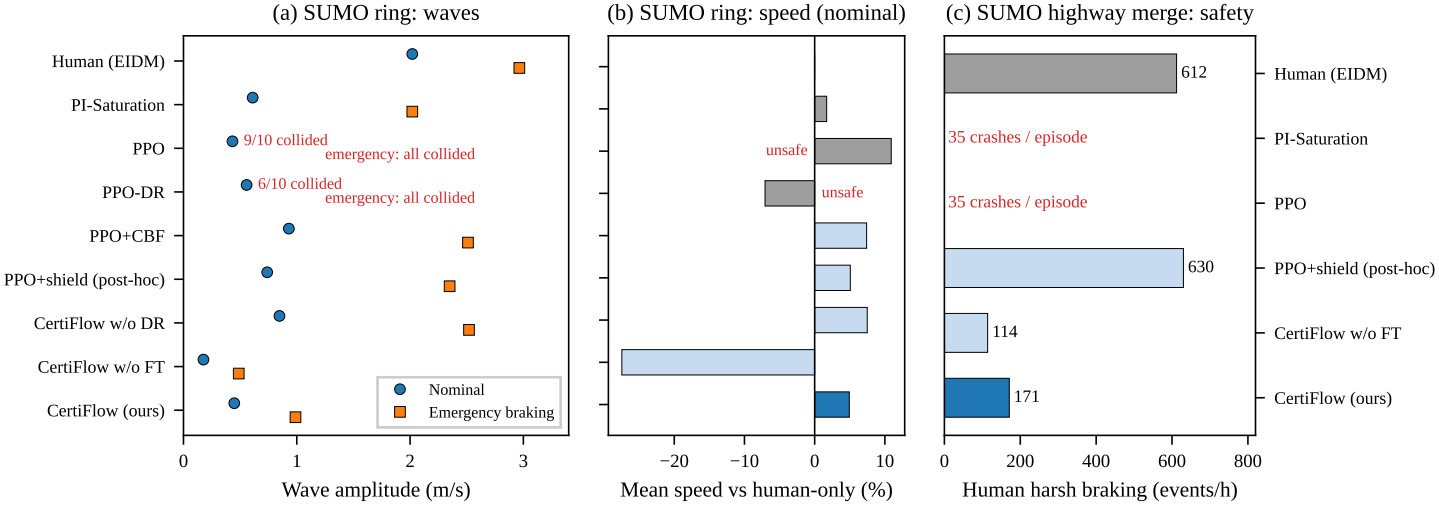
(a) SUMO ring: waves
Human (EIDM)
PI-Saturation
PPO
PPO-DR
PPO+CBF
PPO+shield (post-hoc)
CertiFlow w/o DR
CertiFlow w/o FT
CertiFlow (ours)
9/10 collided
emergency: all collided
6/10 collided
emergency: all collided
Nominal
Emergency braking
0
1
2
3
Wave amplitude (m/s)
(b) SUMO ring: speed (nominal)
unsafe
unsafe
−20
−10
0
10
Mean speed vs human-only (%)
(c) SUMO highway merge: safety
612
35 crashes / episode
35 crashes / episode
630
114
171
Human (EIDM)
PI-Saturation
PPO
PPO+shield (post-hoc)
CertiFlow w/o FT
CertiFlow (ours)
0
200
400
600
800
Human harsh braking (events/h)